Time-Series Analysis of illegal Zama-Zama Mining Expansion

By Thomas RV Elliott

What Is Zama Zama Mining?

- Illegal Artisanal Mining within abandoned mines.
- Often controlled by highly organized criminal syndicates.
- Mostly commonly mining for gold, diamonds or coal

Why is Zama Zama Mining a problem?

• Economic Loss: Illegal mining costs South Africa approximately R70 billion annually in lost revenue, taxes, and royalties.

• Modern Slavery: Syndicates trap vulnerable migrants in debt bondage, forcing them into "underground prisons" for months at a time.

• Organized Crime: Armed gangs protect territories with military-grade weapons, using miners as human shields during police interventions.

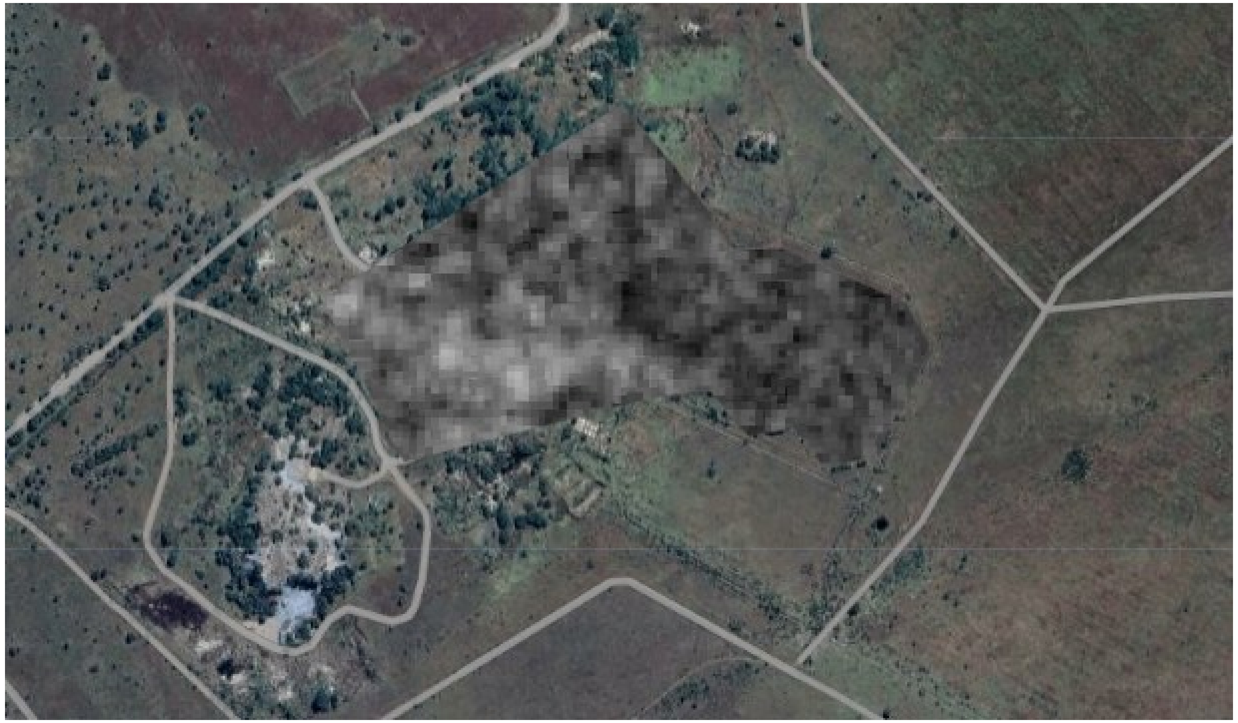
• Environmental Damage: Unregulated operations poison groundwater with toxic mercury and create dangerous sinkholes near communities.

• Fatal Risks: Thousands of miners risk death from gas explosions and rockfalls within the 6,000 abandoned, unsealed mine shafts.

Datasets and Satellites

- Sentinel-1 SAR band
- Sentinel-2 multispectral bands
- Shapefiles used as masks

  where indices fail

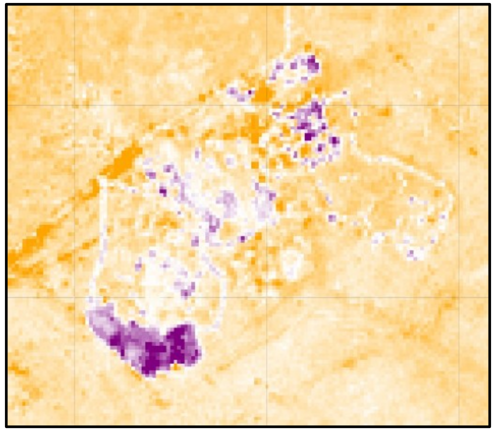


Transformation Techniques

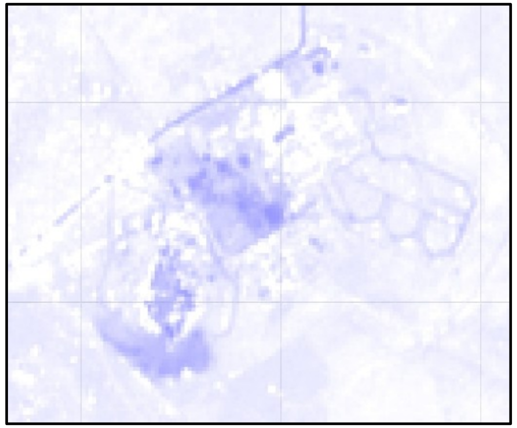
- Spectral Indices
  - NDVI
  - NDWI
  - BAEI
- SAR Backscatter Difference

2017

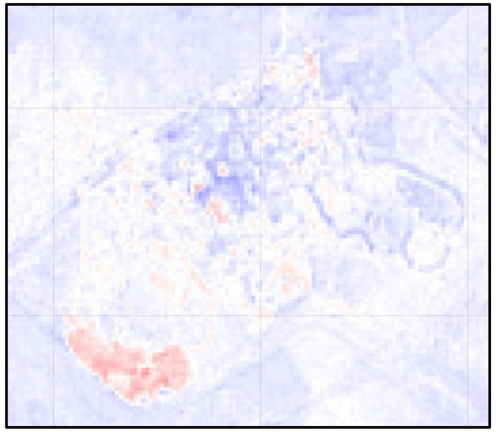
2021



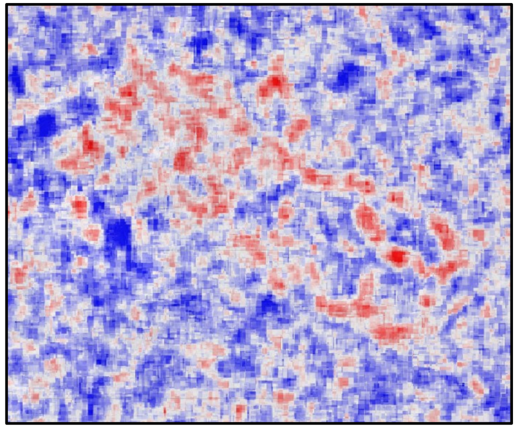
Bitemporal NDVI difference layer (2017 – 2021)



Bitemporal NDWI difference layer (2017 – 2021)



Bitemporal BAEI difference layer (2017 – 2021)



Bitemporal SAR difference layer (2017 – 2021)

Indices Code

```
// Sentinel-2 Optical Base Layers
var s2_2017 = ee.ImageCollection("COPERNICUS/S2_SR_HARMONIZED")
  .filterBounds(studyArea).filterDate('2017-01-01', '2017-12-31')
  .filter(ee.Filter.lt('CLOUDY_PIXEL_PERCENTAGE', 20));

var s2_2021 = ee.ImageCollection("COPERNICUS/S2_SR_HARMONIZED")
  .filterBounds(studyArea).filterDate('2021-01-01', '2021-12-31')
  .filter(ee.Filter.lt('CLOUDY_PIXEL_PERCENTAGE', 20));
```

```
var getSAR = function(startDate, endDate) {
  return ee.ImageCollection('COPERNICUS/S1_GRD')
    .filterBounds(studyArea)
    .filterDate(startDate, endDate)
    .filter(ee.Filter.eq('instrumentMode', 'IW'))
    .filter(ee.Filter.listContains('transmitterReceiverPolarisation', 'VV'))
    .select('VV')
    .median()
    .clip(studyArea);
};
```

```
var getNDVI = function(image) {
  return image.normalizedDifference(['B8', 'B4']).rename('NDVI');
};
```

```
var ndwi2021 = s2_2021.map(function(image) {
  return image.normalizedDifference(['B3', 'B8']).rename('NDWI');
}).median().clip(studyArea);
```

```
var getBAEI = function(image) {
  var baei = image.expression(
    '(RED + 0.3) / (GREEN + SWIR1)', {
      'RED': image.select('B4'),
      'GREEN': image.select('B3'),
      'SWIR1': image.select('B11')
    }).rename('BAEI');
  return image.addBands(baei);
};
```

```
var sar2017 = getSAR('2017-01-01', '2017-01-31');
var sar2021 = getSAR('2021-01-01', '2021-01-31');
var sarDiff = sar2017.subtract(sar2021);
```

Create Diff Layer

```
var zamaZamaLikelihood = sarScore.add(baeiScore).add(ndviStableLow).clip(studyArea);
```

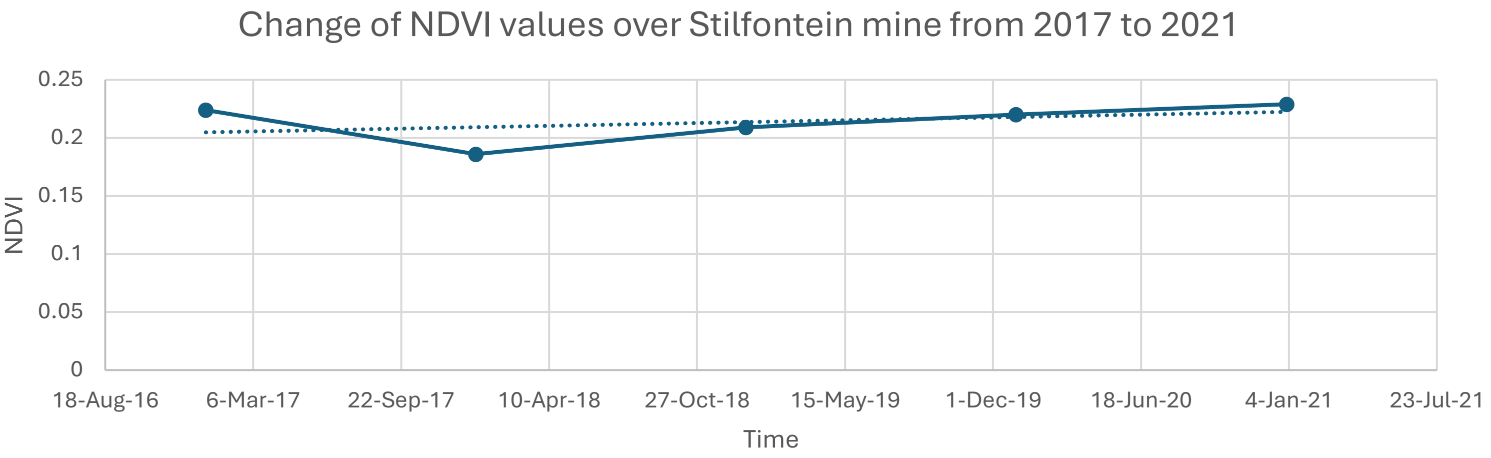
```
var maskWaterShapefile = ee.FeatureCollection('projects/git712prac/assets/nospacesriversanddams');

var maskImage = ee.Image(1).paint(maskWaterShapefile, 0);
```

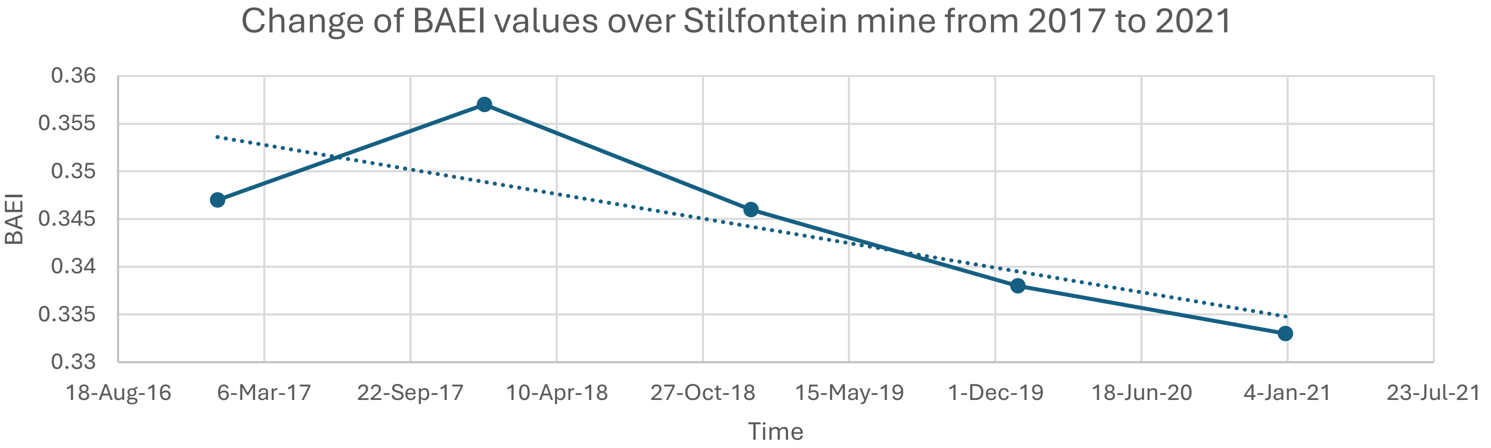
The Method

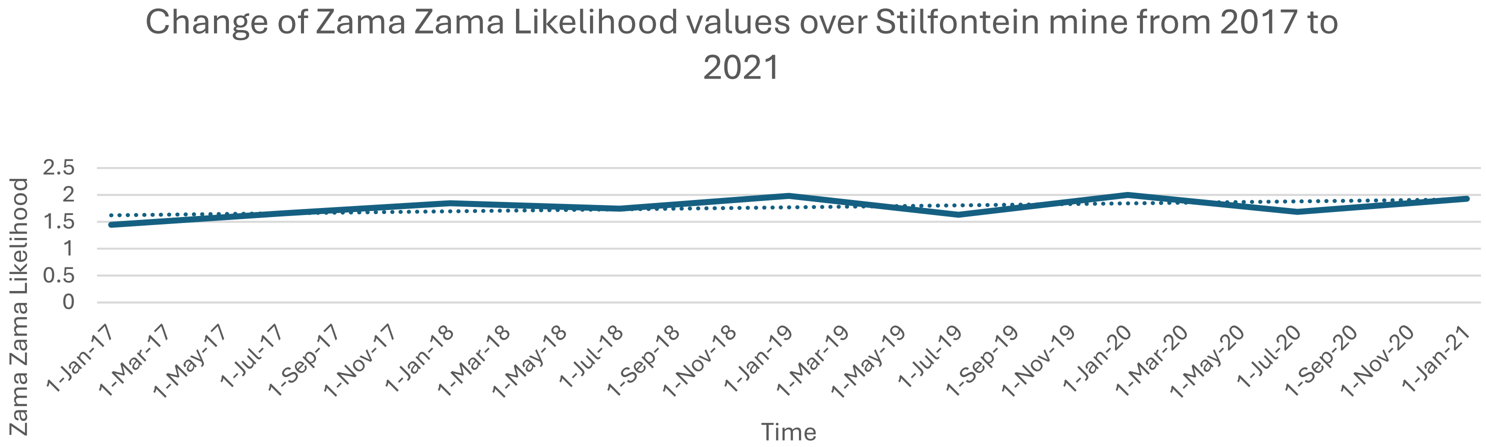
1. **Acquire Base Imagery:** Retrieve Sentinel-1 SAR (VV backscatter) and Sentinel-2 optical imagery for the 2017 baseline and the 2021 analysis period.
2. **Execute Data Transformations:** Calculate the Normalized Difference Vegetation Index (NDVI), Normalized Difference Water Index (NDWI), and Built-Up Area Extraction Index (BAEI).
3. **Generate Difference Layers:** Subtract the 2017 SAR and BAEI values from the 2021 values to isolate areas of structural change and infrastructure loss over time.
4. **Build Environmental Masks:** Isolate pixels with water signatures (using NDWI) and healthy vegetation (using NDVI) to create a noise-reduction mask.
5. **Filter False Positives:** Apply the environmental masks to the SAR difference layer to strip out irrelevant backscatter changes caused by farming or fluctuating dam levels.
6. **Assign Threshold Scores:** Apply heuristic thresholds to flag high-risk pixels, assigning a score of 1 for significant SAR structural change, a score of 1 for a drop in the building index, and a score of 1 for sustained bare earth.
7. **Calculate Final Index:** Sum the threshold scores to generate the final Zama Zama Likelihood Index, resulting in a mapped classification ranging from 1 (low likelihood) to 3 (high likelihood).

Time Series Analysis

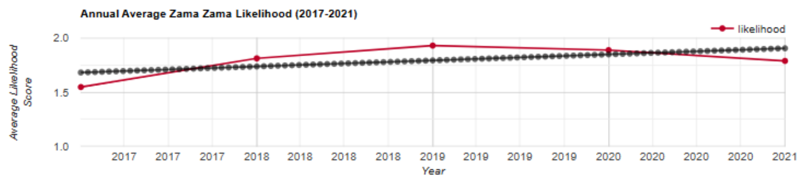
- NDVI decreases at the same time that BAEI increases.
- NDVI shows an overall increase in vegetation within the region
- VV backscatter remains fairly constant
- VV backscatter peaks in early 2018 and early and late 2019





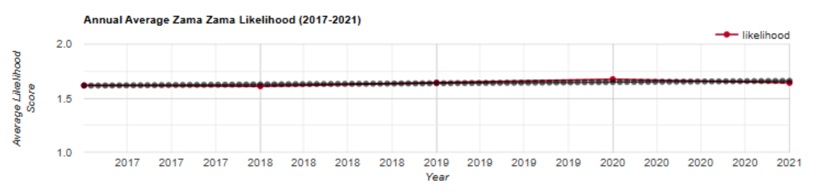


Results

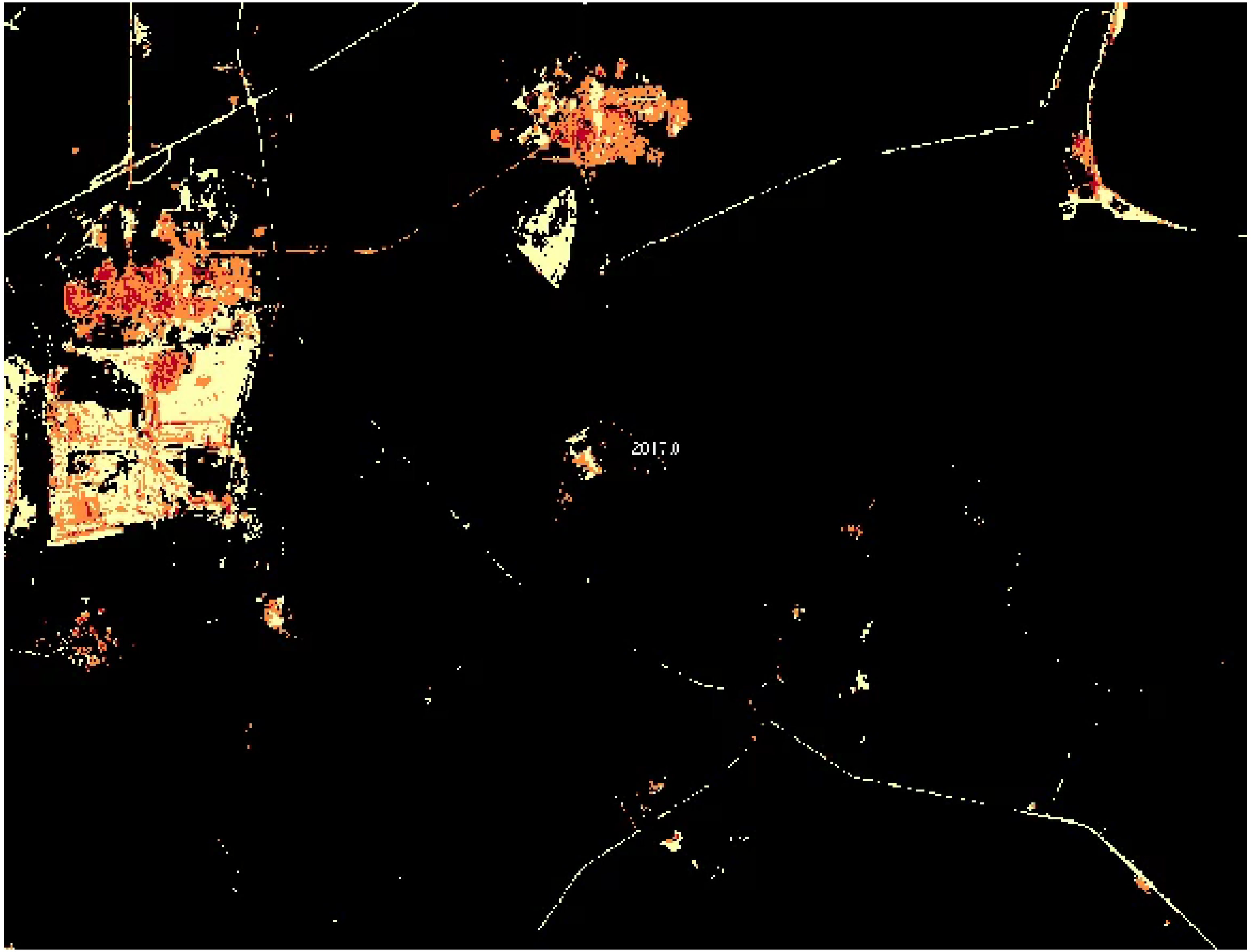


- The results show an increase of likely zama zama mining disturbance from 2017 to 2019 and then a slight decrease to 2021.
- These results lineup with the known timeline of zama zama mining within the area.
- The use of a mean or a median image limits the accuracy.
- The decrease in likelihood is deceiving.

Model Reliability

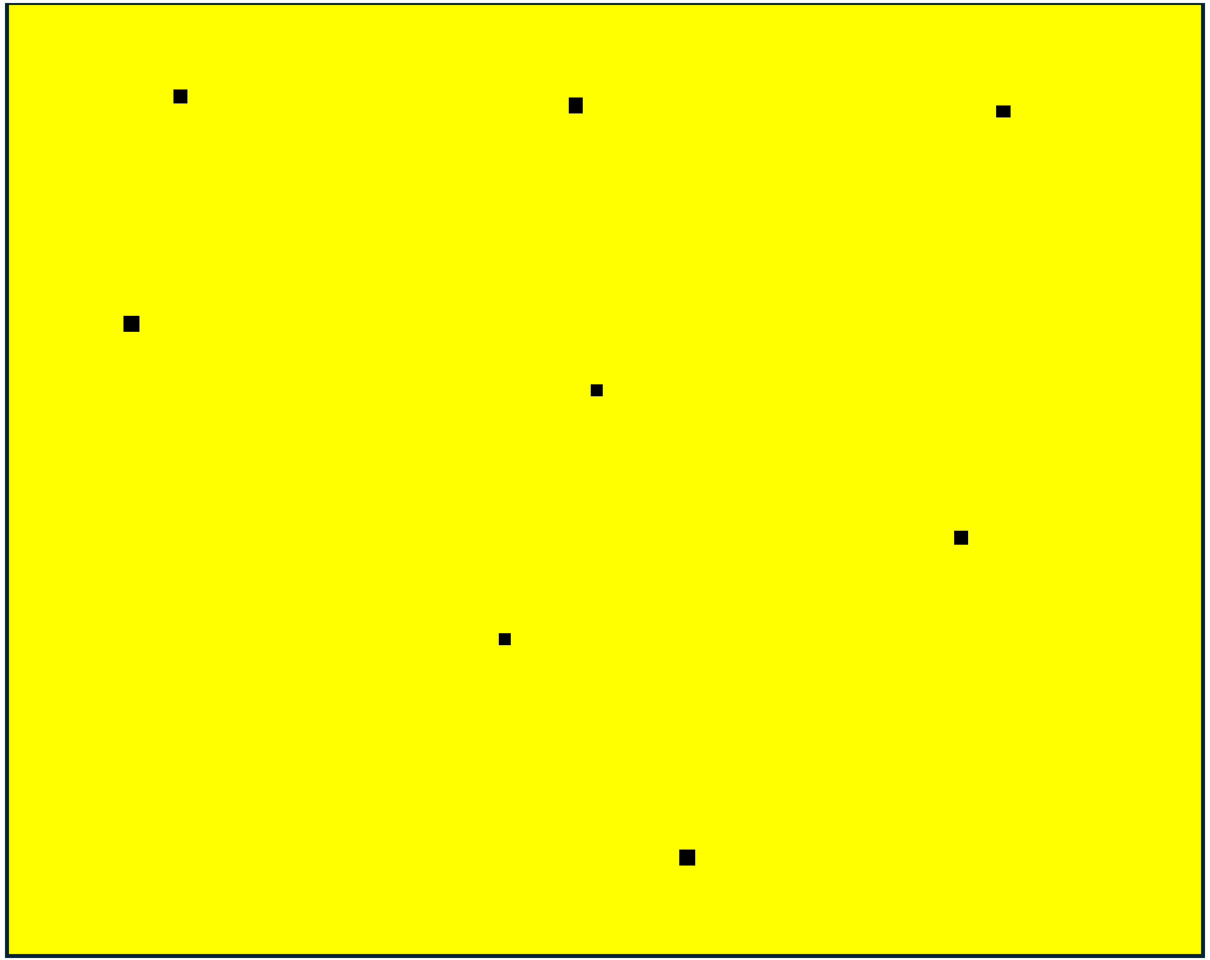


- Model correctly identifies areas of mining.
- Model does identify areas of abandoned mines as medium risk
- Confused some active mines with zama zama mining
- Could be improved with a dataset of illegal / legal mines.

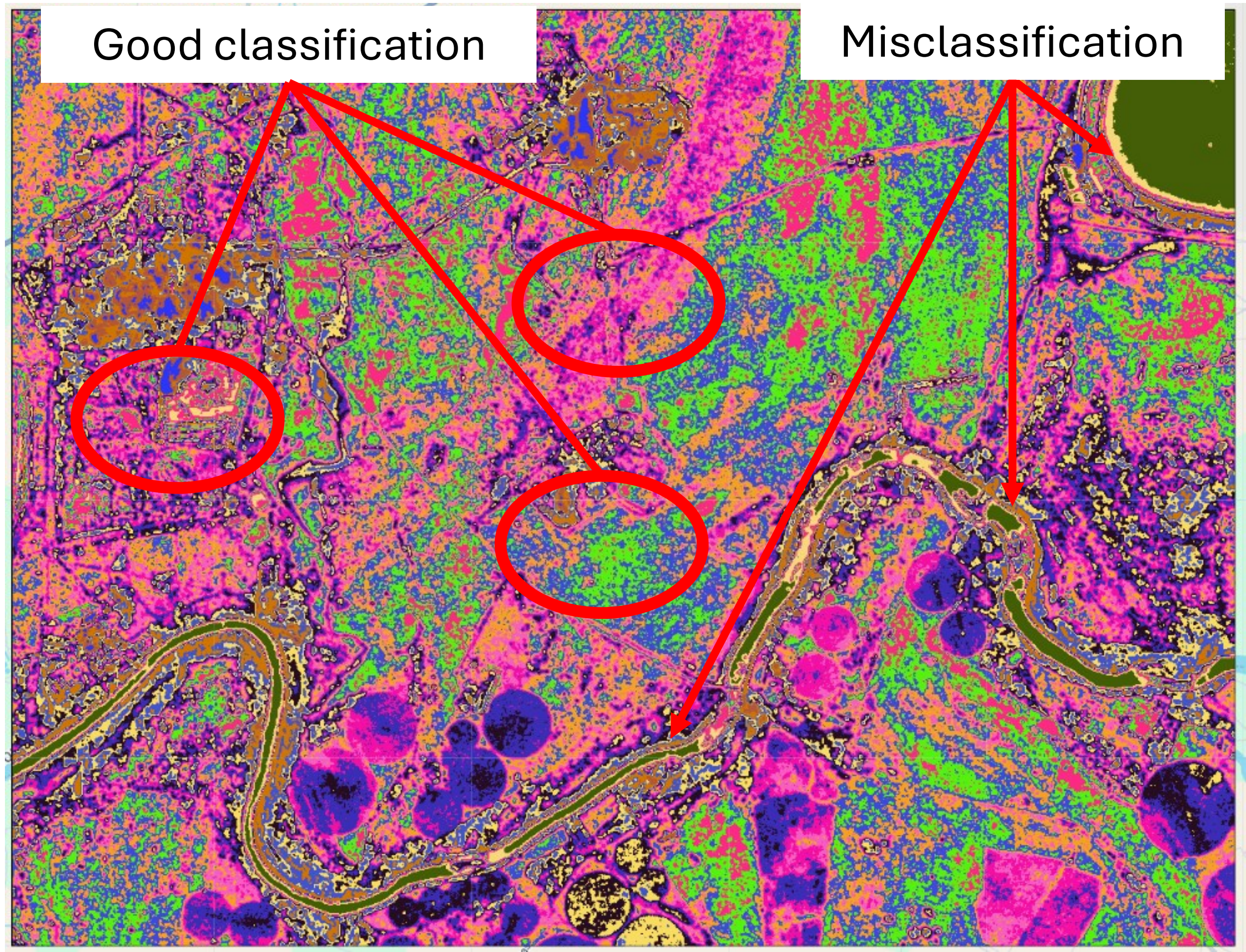


Classifications

Supervised Classification



Unsupervised Classification



Classification Recommendations

- Supervised Classification
  - Make more classes
  - Ensure large classes that could be confused are separated
  - Use a larger training dataset
  - Use many small areas for training data
- Unsupervised Classification
  - Make the 'Training data' smaller and more precise

```
// Define a region in which to generate a segmented map.
var region = ee.Geometry.Polygon(
  [[[26.789431113062967,-26.95601316922397],
    [26.893801230250467,-26.95601316922397],
  [26.893801230250467,-26.884840511831076],
    [26.789431113062967,-26.884840511831076],
    [26.789431113062967,-26.95601316922397]]]);

// Load a Landsat composite for input.
var input = ee.ImageCollection('COPERNICUS/S1_GRD')
  .filterBounds(region)
  .filterDate('2017-01-01', '2021-01-01')
  .filter(ee.Filter.eq('instrumentMode', 'IW'))
  .filter(ee.Filter.listContains('transmitterReceiverPolarisation', 'VV'))
  .select('VV')
  .median()
  .clip(region);

// Display the sample region.
Map.setCenter(26.838021312101763,-26.918881716137655, 16);
Map.addLayer(ee.Image().paint(region, 0, 2), {}, 'region');

// Make the training dataset.
var training = input.sample({
  region: region,
  scale: 10,
  numPixels: 5000
});

// Instantiate the clusterer and train it.
var clusterer = ee.Clusterer.wekaKMeans(15).train(training);

// Cluster the input using the trained clusterer.
var result = input.cluster(clusterer);

// Display the clusters with random colors.
Map.addLayer(result.randomVisualizer(), {}, 'clusters');
```

Conclusion

- The principle of identifying illegal mining is plausible.
- SAR data should be used primarily for identification while multispectral data should be used primarily for masking noise.
- It should be investigated more in depth.
- For future research, more training data should be used.
- Using SLC Coherence layers should be tested.